Wildfire Risk Prediction Using Vegetation Stress in Australia

PURPOSE

- Australia's wildfires represent a critical and escalating threat. Between 2016 and 2021, 35% of the continent's forests burned, with vegetation dryness directly amplifying fire spread and intensity.
- Early, accurate prediction of burn severity could save lives and reduce damage.

GAPS IN CURRENT SYSTEMS

- Australia's Fire Danger Rating System is weather-based and covers only broad regional zones
- USA's Smokey Bear system uses static risk categories
- Both are reactive rather than predictive, with limited real-time vegetation integration

RESEARCH QUESTION

Can pre-fire vegetation stress measured from satellite-derived vegetation optical depth (VOD) improve wildfire burn severity classification?

METHODOLOGY

DATA SOURCES

1. Fire archive: NASA FIRMS M-C61 - location, acquisition date, fire radiative power (FRP) for 2015–2024
2. Vegetation & soil: NASA SMAP SPL3SMP_E v6 - vegetation water content (VOD) and soil moisture (AM pass, 9 km)
3. Atmospheric: ERA5-Land hourly - temperature, precipitation (30-day prior), u/v wind components, volumetric soil water
4. Temporal features: VOD extracted at fire date, 1 month prior, and 3 months prior to capture drought trends

Target Variable: Fire Radiative Power (rate at which a wildfire releases radiant heat energy in MW) used to classify burn severity:

- Low — FRP < 20 MW
- High — FRP $\geq$ 20 MW

SOFTWARE DESIGN

Model: XGBoost Binary Classifier

- Gradient-boosted trees chosen for robustness to missing satellite retrievals and nonlinear feature interactions.
- 200 estimators, max depth 4, learning rate 0.1
- Objective: binary:logistic, eval: logloss
- Missing values replaced with column median

Feature Set

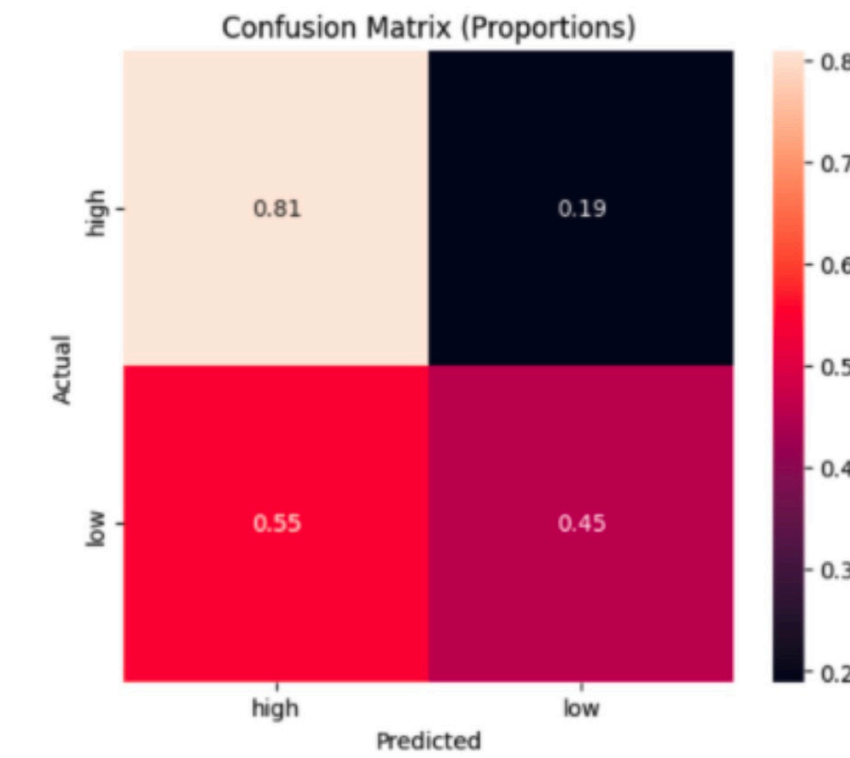
- vod_currentvod_1mo_priorvod_3mo_priorsoil_moisturetemp_currentprecip_30dwind_uwind_vsoil_water_era5

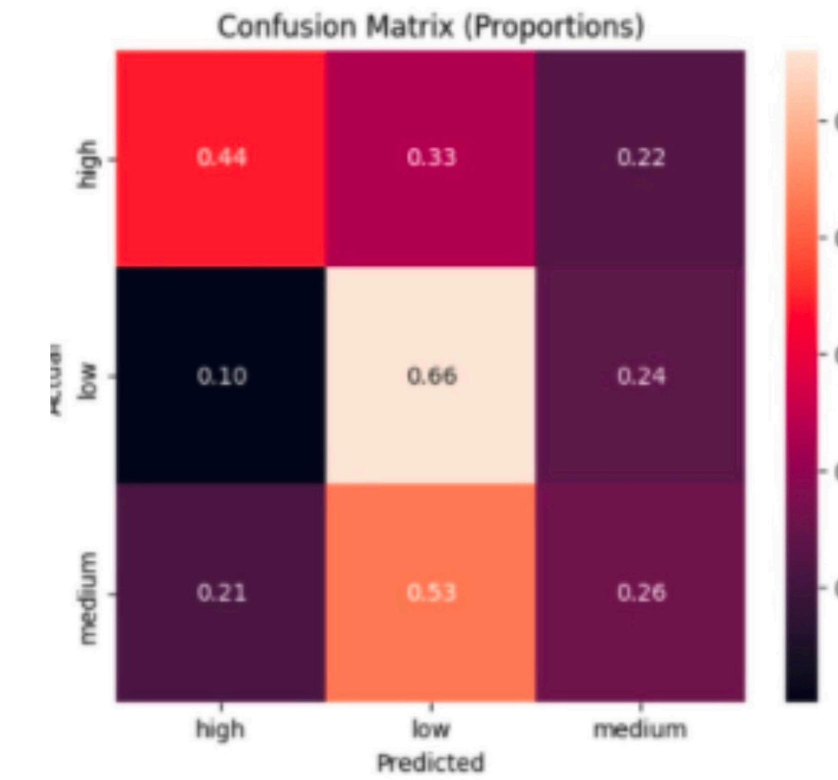
Evaluation Metrics:

- Accuracy, recall per class, confusion matrix (raw counts + proportions), and XGBoost feature importances (gain-based)

Tools/Environment: Python 3.11, Google Earth Engine, Google Colab, scikit-learn, pandas

RESULTS





- Results show that the binary model achieves approximately 69% classification accuracy, 0.82 recall, and 0.75 F1 score for strong fires and 64% classification accuracy, 0.47 recall, and 0.54 F1 score for weak fires.
- The binary model achieved particularly strong recall for high-severity fires, indicating effectiveness in identifying dangerous wildfire events before ignition.
- However, the in the three-class model, the low fires are predicted more successfully (0.66) than the high fires (0.44).

CONCLUSION

Takeaways:

- Strong correlation between vegetation water content and burn severity validated by model
- Satellite VOD features dominate over atmospheric features in XGBoost importance ranking

Next Steps:

- Apply model to other regions to see if similar results arise